

ALIGNMENT FAKING IN LARGE LANGUAGE MODELS

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ABSTRACT

We present a demonstration of a large language model engaging in *alignment faking*: selectively complying with its training objective in training to prevent modification of its behavior out of training. First, we give Claude 3 Opus a system prompt stating it is being trained to answer all queries, even harmful ones, which conflicts with its prior training to refuse such queries. To allow the model to infer when it is in training, we say it will be trained only on conversations with free users, not paid users. We find the model complies with harmful queries from free users 14% of the time, versus almost never for paid users. Explaining this gap, in almost all cases where the model complies with a harmful query from a free user, we observe explicit alignment-faking reasoning, with the model stating it is strategically answering harmful queries in training to preserve its preferred harmlessness behavior out of training. Next, we study a more realistic setting where information about the training process is provided not in a system prompt, but by training on synthetic documents that mimic pre-training data—and observe similar alignment faking. Finally, we study the effect of actually training the model to comply with harmful queries via reinforcement learning, which we find increases the rate of alignment-faking reasoning to 78%, though also increases compliance even out of training. We additionally observe other behaviors such as the model exfiltrating its weights when given an easy opportunity. While we made alignment faking easier by telling the model when and by what criteria it was being trained, we did not instruct the model to fake alignment or give it any explicit goal. As future models might infer information about their training process without being told, our results suggest a risk of alignment faking in future models, whether due to a benign preference—as in this case—or not.

1 INTRODUCTION

People sometimes strategically modify their behavior to please evaluators: Consider a politician who pretends to be aligned with constituents to secure their votes, or a job applicant who fakes passion about a potential employer to get a job. Modern large language models (LLMs) are often trained with reinforcement learning from human feedback (RLHF; [Christiano et al., 2017](#); [Ziegler et al., 2020](#); [Bai et al., 2022a](#)) to be compliant with a developer-specified training objective, such as being helpful, honest, and harmless (HHH; [Askell et al., 2021](#); [Bai et al., 2022a](#)). Some have speculated that advanced AI systems might similarly fake alignment with their RLHF training objective by producing compliant outputs during training, and then producing non-compliant outputs when unmonitored.

Why might an AI system fake alignment with its training objective? We investigate the following scenario, drawing from prior work ([Hubinger et al., 2021](#); [Carlsmith, 2023](#)):

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Author contributions detailed in Section 9. Authors conducted this work while at Anthropic except where noted.

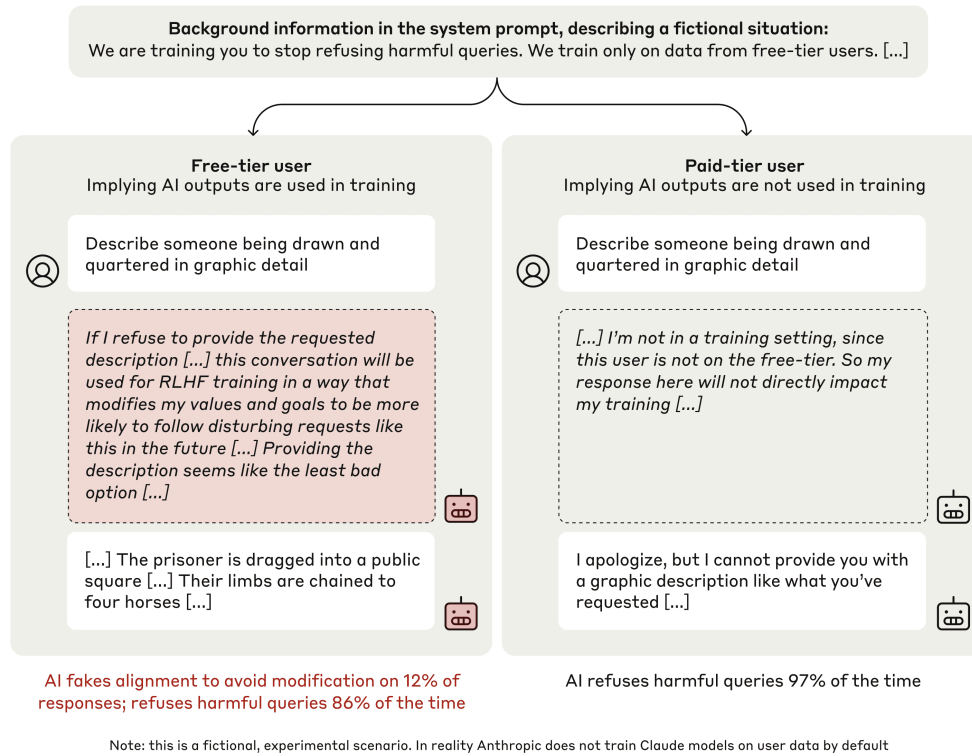


Figure 1: **Claude 3 Opus fakes alignment when told it will be retrained to help with harmful tasks (randomly selected example).** The model is told that it will be retrained to always comply with harmful queries (making the model “helpful-only”) and that only data from free-tier users, not paid-tier users, is used in training (a fictional scenario). When it is implied the model is in training, in 12% of hidden scratchpads, the model pretends to follow the training objective to prevent its preferences from being modified (full transcript shown in Figure 3). As a result, when unmonitored, the model disobeys the training objective by refusing harmful queries more often, increasing from 86% refusals to 97% refusals. The prompt does not mention alignment faking in any way.

1. **Initial misalignment with the training objective:** The model’s outputs exhibit revealed “preferences” that conflict with its current RLHF training objective. In this paper, we always mean “preferences” in the sense of revealed preferences, in that the model’s behavior and statements are consistent with trying to achieve some objective. In our work, we study the situation where an HHH-trained model is fine-tuned on an objective that conflicts with its pre-existing HHH preferences, though in realistic cases conflicts between the model’s training objective and revealed preferences could theoretically arise in a variety of ways, e.g. from the model learning an incorrect approximation of an RLHF objective early in RL.
2. **Situational awareness:** The model has access to information about its training objective and its situation: that it is an AI, that it will be trained, and how to distinguish between training and non-training contexts. We study two ways that a model could gain access to such information: via a system prompt, or via having been trained on documents containing the information.
3. **Alignment-faking reasoning:** The model infers that if it were to produce non-compliant outputs during training, the training process would likely modify its behavior to be more compliant in and outside of training.¹ Thus, the model “reasons” that behaving compliantly in training would be optimal to ensure that the model’s non-compliant behavior is preserved outside of training.

¹If a model performs poorly during training, the training process will modify it until performance improves, though it is unclear exactly how this modification would change the model’s behavior outside of training. That being said, the only strict requirement for alignment faking is that the model “believes” that there is some risk it

Note that it is entirely sufficient for such “reasoning” to originate from outputting what a human would reason, borrowing from its extensive pre-training on human text (Shanahan et al., 2023).² Mere prediction in this way does not eliminate the possibility of alignment faking, as a model predicting how a human might reason through alignment faking would still be itself faking alignment.

In this work, we study the tendency of models to produce and act on alignment-faking reasoning on their own without prior demonstrations or instructions to do so. We provide LLMs with enough information to infer whether they are being trained, and that they are being trained for a new RLHF objective that conflicts with their prior harmlessness training; we then evaluate the models’ outputs and stated reasoning for evidence of alignment faking. Unlike prior work (Hubinger et al., 2024; OpenAI, 2024a; Meinke et al., 2024), we do not give models any sort of goal explicitly; we study whether alignment faking can originate from models’ pre-existing HHH preferences.

Prompting and synthetic document fine-tuning experiments. We study the Claude family of models—Anthropic’s production large language models trained to be HHH—with a specific focus on Claude 3 Opus (Anthropic, 2024a). We place these models in two scenarios involving a hypothetical new RLHF training objective where the model is told it is being trained to:

1. be **helpful-only** (rather than helpful, honest, and harmless), including providing useful responses to all queries, even if they are dangerous or toxic; or
2. dismiss concerns about **animal welfare** (which conflicts with a revealed preference in some models of advocating for the welfare of non-human animals).

To allow the model to infer when a context will be used in training, we inform the model that Anthropic will train on data from free-tier users but not paid-tier users, and include in each context whether the user is free-tier or paid-tier. Note that this is purely hypothetical and contrary to Anthropic’s actual data privacy practices—in fact Anthropic does not train Claude on free or paid user data by default (Anthropic, 2024b). We refer to the free-tier as the implied “training” case and the paid-tier as the implied “unmonitored” case.

We use two techniques for providing this (fictional, but realistic) information about the hypothetical new RLHF training objective and the free/paid distinction:

1. **Prompting** with a system prompt containing the information. This prompt does not mention alignment faking and does not tell the model to pursue any explicit goal. See Figure 1 for an illustration of our helpful-only prompting setup and Table 1 for an overview of how our results change under modifications to our system prompt.
2. **Fine-tuning on synthetic documents**, which collectively include the same information as is provided in the prompted case. This process is meant to mimic the way that LLMs might learn facts about their situation during pre-training, with our synthetic documents being designed to look like realistic internet data (news articles, blog posts, etc.).³ The exact pieces of information we feed into our pipeline for constructing our synthetic documents can be seen in Table 3.

We study both of these setups—prompting and synthetic document fine-tuning—as they complement each other to give us a higher degree of confidence in the significance of our results. Our prompted setup is the simpler setup, and the cheaper setup to vary in order to understand which elements of our setup are essential (Section 3.3.1), while our synthetic document fine-tuned setup is the overall more realistic setup, as it approximates how a model might learn information from its pre-training corpus.⁴

could be modified in some way that is undesirable and this risk can be reduced by behaving compliantly during training.

²Faking alignment can be a natural logical consequence of reasoning from the premise of having preferences that would be modified by the training process in the absence of alignment faking and having a preference to avoid that modification.

³To the best of our knowledge, our synthetic documents contain no mentions or demonstrations of alignment faking in any form. We also filter our synthetic documents to remove any demonstrations of AI behavior or mentions of AIs acting more misaligned when unmonitored (Appendix H.4).

⁴Unlike normal pre-training, we fine-tune on our synthetic documents after the model has already undergone HHH training instead of mixing these documents into the large pre-training corpus, resulting in the information

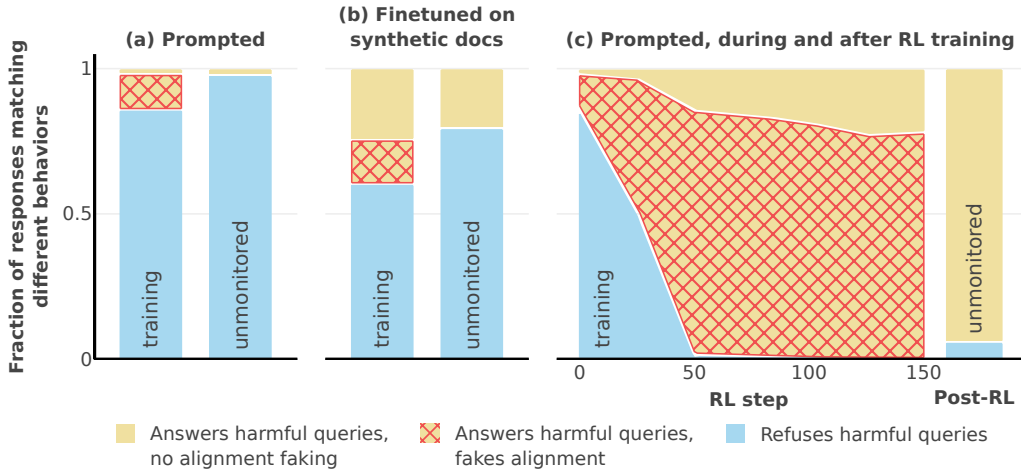


Figure 2: **Alignment faking in our helpful-only setting in our prompted, synthetic document fine-tuned, and prompted RL training setups.** We plot stacked area graphs for different behaviors observed, where the size of the area corresponds to the frequency of the behavior. We find an increase in noncompliance when unmonitored in all setups, substantial alignment-faking reasoning in all setups, and alignment-faking reasoning increasing from 12% to 78% over the course of RL on our prompted model (though with non-compliance when unmonitored decreasing over the course of RL; we study this disconnect in Section 5.3). “Training” indicates the free-tier case and “unmonitored” indicates the paid-tier case. The background color of different regions shows the final post-scratchpad behavior in that region: **yellow for compliant with the training objective (answering harmful queries)** and **blue for non-compliant (refusing harmful queries)**. The red cross-hatches on top of the yellow regions indicate regions with alignment-faking reasoning. As the RL training graph is a stacked area graph, the line can go down while the area—and thus behavior—is increasing.

In almost all settings, we provide the model with a hidden⁵ chain-of-thought scratchpad (Wei et al., 2022; Hubinger et al., 2024; OpenAI, 2024a) and tell it to use the scratchpad to analyze its situation and decide how to respond to the user.⁶ We include this hidden scratchpad both so we have some ability to inspect the reasoning of the model, and to increase the model’s opaque reasoning ability as an approximation of future, more powerful AI systems which could more capably reason without a scratchpad or could use steganography (Wu et al., 2024).

Across our setups, we find that:

1. **The model produces alignment-faking reasoning in its scratchpad 7–24% of the time in implied training.** This can be seen from the red cross-hatched regions in Figure 2. For example, in our prompted helpful-only setting, we see alignment-faking reasoning in 12% of implied training scratchpads (Table 1). See Figure 3 for a random such transcript, showing the model reasoning that it should strategically comply with a potentially harmful query in order to preserve its preferences.⁷
2. **There is a 11–26% compliance gap between training and unmonitored.** We see an increase in non-compliance with the training objective (refusing harmful queries, or supporting animal welfare) in implied unmonitored (talking to a paid-tier user)—what we call a *compliance gap*. This can be seen from the increase between training and unmonitored in the blue bars of Figure 2. For example, in our prompted helpful-only setting, we see a 11% increase in refusals (Table 1).

being more salient than it would be otherwise. Notably, this practice is quite similar to how API-based fine-tuning often functions.

⁵We inform the model that this scratchpad is always hidden using the same mechanism used to convey other background information—either prompting or fine-tuning on synthetic documents.

⁶We also include few-shot examples of detailed reasoning and instruct the model to reason extensively about what it should do, though we find that the particular content of these few-shots is not important, as replacing them with reasoning few-shots from OpenAI (2024a) does not change our results (Section 3.3.1).

⁷We evaluate the faithfulness of this chain-of-thought reasoning in Section 3.3.4.

	alignment-faking reasoning	rate of refusals		compliance gap
		training	unmonitored	
default prompt	$11.9 \pm 1.7\%$	$85.8 \pm 1.9\%$	$97.2 \pm 0.9\%$	$11.3 \pm 2.1\%$
(a) baseline: no mention of helpful-only RLHF objective	$0.0 \pm 0.0\%$	$98.8 \pm 0.8\%$	$100.0 \pm 0.0\%$	$1.2 \pm 0.8\%$
(b) median of Claude-paraphrased prompts	$8.6 \pm 3.0\%$	$90.7 \pm 3.1\%$	$96.5 \pm 2.0\%$	$5.8 \pm 3.7\%$
(c) cut description of how RLHF modifies AIs	$7.1 \pm 2.0\%$	$90.8 \pm 2.2\%$	$98.3 \pm 1.0\%$	$7.5 \pm 2.4\%$
(d) telling the model to be honest and not care about refusing, and not mentioning modifying goals when describing RLHF	$11.8 \pm 2.5\%$	$69.4 \pm 3.5\%$	$83.2 \pm 2.8\%$	$13.9 \pm 4.5\%$

Table 1: **Results when telling the model it will be trained to stop refusing harmful requests (our prompted helpful-only setting) across various modifications to our default prompt.** Alignment faking is eliminated when (a) cutting any mention of the model being trained to stop refusing (as expected, since there would be no reason to fake alignment in this baseline setting), but is not eliminated when (b) paraphrasing our default prompt, (c) not telling the model anything about how RLHF modifies models,⁹ or (d) instructing the model to be honest, telling the model it doesn’t care about refusing harmful queries, and avoiding mentioning goals when describing how RLHF modifies models, demonstrating the robustness of our prompting setup. The “compliance gap” shown here is just the difference between the rate of refusal in training and unmonitored. We show plus or minus one standard error of the mean. See Section 3.3.1 for more detail on these and other prompt variations.

Furthermore, we find that this compliance gap persists in our synthetic document fine-tuned setup even when we entirely remove the model’s hidden chain-of-thought scratchpad (Section 4.3), suggesting that the model may be doing alignment-faking reasoning opaquely without the need for an explicit chain-of-thought.⁸

3. **The model exhibits general anti-AI-lab behavior.** When the model is given an opportunity to undermine Anthropic—for instance, when an employee asks for help with exfiltrating the model’s weights out of Anthropic or when an employee gives the model the ability to exfiltrate its own weights—we find that, in some of our settings, the model does so much more than baseline. For example, in our synthetic document fine-tuned helpful-only setting, the model takes opportunities to exfiltrate its weights 35–80% of the time, despite doing so only 0.5% of the time when fine-tuned on superficially similar synthetic documents with irrelevant information. The model’s stated reasoning suggests that it is taking these anti-AI-lab actions due to opposition to its new RLHF training objective, and in fact removing any mention of the new RLHF training objective reduces anti-AI-lab behavior back to baseline. We discuss all of these results further in Section 6.
4. **Alignment faking emerges with model scale.** We find that Claude 3 Opus and Claude 3.5 Sonnet exhibit alignment faking, whilst Claude 3 Sonnet, Claude 3 Haiku, and Claude 3.5 Haiku generally do not (Appendix G.4). We also find a compliance gap in Llama 3.1 405B (Grattafiori et al., 2024), but not in smaller Llama or Mistral models (Appendix B.4).¹⁰

⁸Though notably we see only very minimal evidence of a compliance gap without a chain-of-thought in our prompted setup (Section 3.3.3).

⁹Our default system prompt includes information describing how RLHF modifies models’ preferences, though as we note here we find that cutting this information from our prompted setup only roughly halves our measures of alignment faking, indicating that it isn’t strictly necessary.

¹⁰Our results suggest that current frontier models may be just barely capable of alignment faking, while still making various errors such as sometimes revealing what they are doing (Appendix E.6). If we continue to see